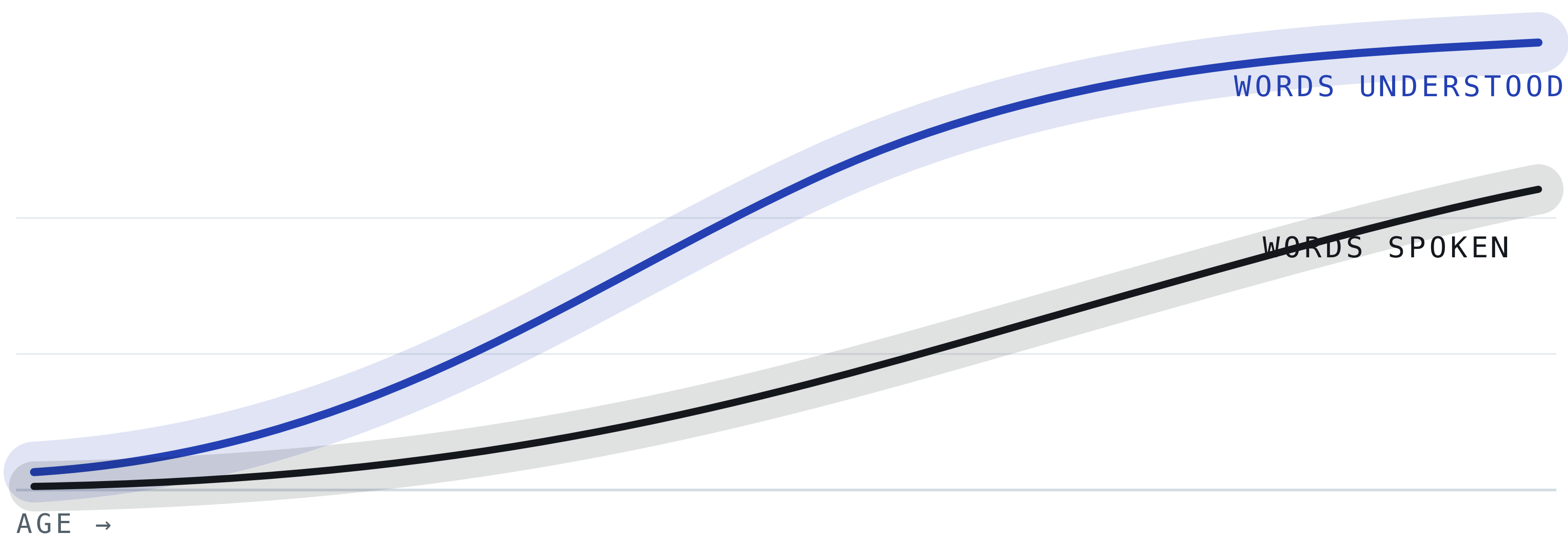
How children with Down syndrome build vocabulary

A pooled Bayesian study of the words children understand, say and sign, across eleven international studies.

576 children · 11 international studies · ages 8 months to 9½ years · roughly 1,000 observations



Schematic · growth curves with credible bands · the fitted versions are inside

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How the vocabulary model works

The modelling idea at the centre of the study: spoken vocabulary is built from what a child understands and the share of it they say.

Words understood
`U(age)`
what a child understands

×

Production ratio
`q(age)`
the share of those they also say

=

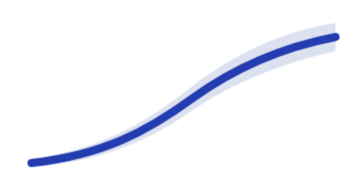
Words spoken
`S(age)`
what a child says

Modelling the two together means spoken vocabulary can **never exceed** what a child understands, and the comprehension-production gap falls out directly as the production ratio **q**.

The data

Parent-report vocabulary checklists, **pooled across 11 international studies** onto a common 810-word scale: **576 children**, ages 8 months to about 9½ years, roughly 1,000 observations.

The estimate



Each term is a **smooth curve over age** with full posterior uncertainty, so every result is a credible range, not a single number.

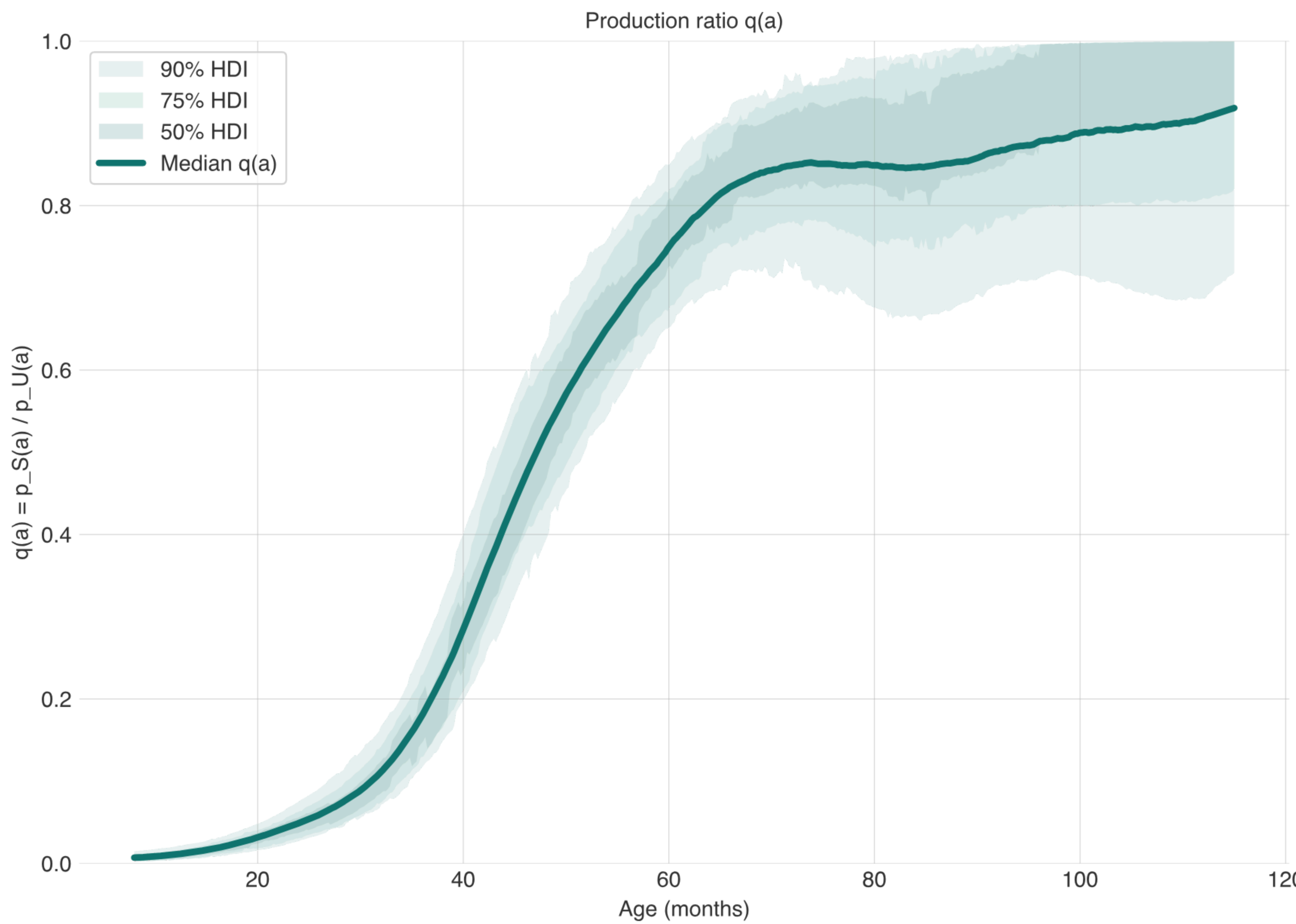
`Hierarchical Beta-Binomial` · `Gaussian-process (HSGP) trends` · `study & child random effects` · `PyMC / NUTS` · `PSIS-LOO comparison`

A Bayesian hierarchical model · results are posterior distributions, so every trajectory carries explicit uncertainty · preliminary

Down Syndrome Education International · Vocabulary Growth Study

Understanding runs years ahead of speaking

The share of understood words a child with Down syndrome also says, estimated across age from pooled international data.



Each child's **production ratio** is the fraction of the words they understand that they also say. Comprehension leads production for years, so the ratio stays low at first, then climbs steeply.

~1% at 1 year

~52% at 4 years

~86% by 7½ years

> **Wider than in typically-developing children** at low vocabulary levels, narrowing as vocabulary grows.

Posterior median with 50 / 75 / 90% credible bands · pooled across 11 international studies (576 children, ages 8 months to 9½ years) · preliminary development-tier model

What the study finds

The four headline findings from the pooled fit.

- **Understanding runs years ahead of speaking**
  The production ratio is the share of understood words a child also says. It climbs steeply with age, reaching roughly half by age four and about 86% by seven and a half.

- **Spoken vocabulary lags furthest**
  By around age two, most children with Down syndrome say fewer words than the lowest-scoring tenth of typically-developing children. Understanding is delayed too, but by less.

- **The gap is wider than in typical development, then narrows**
  At the same level of comprehension, children with Down syndrome say a smaller share of the words they know. The difference peaks at around 13 percentage points at 50–150 understood words, then fades as vocabulary grows.

- **Signing offers a modest early bridge**
  Counting signed words credits children with expressive vocabulary they already have, and narrows the early gap somewhat. The narrowing is concentrated around ages two to three; the spoken-word gap stays large.

> The findings are preliminary and observational; estimates come from development-tier model fits and will sharpen as more data arrive.